

APOORVA BANDI

AI/ML Engineer | bandiapoorva1@gmail.com | +1 682-373-3197

PROFESSIONAL SUMMARY

- AI/ML Engineer with 5+ years delivering production-grade machine learning across healthcare and financial transaction environments using Python, SQL, PySpark, Java, Azure, and AWS.
- Designed classification, forecasting, NLP, document classification, fraud detection, and anomaly solutions with Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, spaCy, autoencoders, Prophet, and ARIMA.
- Built ML scoring APIs and microservices with FastAPI, Flask, Spring Boot, AWS Lambda, and ECS, reducing p95 response latency by 30% across production systems.
- Architected ETL, batch, and streaming workflows with Azure Data Factory, Databricks, Delta Lake, Kafka, Spark, Snowflake, and Airflow, processing 10M+ records daily.
- Automated reproducible training, deployment, monitoring, experiment tracking, artifact versioning, and governance with MLflow, Azure ML, Terraform, Azure DevOps, Docker, and CI/CD.
- Implemented Generative AI with Azure OpenAI, OpenAI API, LangChain, LlamaIndex, Hugging Face Transformers, LoRA/QLoRA, vLLM, quantization, summarization, and synthetic data generation.
- Developed RAG systems with Azure Cognitive Search, vector embeddings, semantic search, pgvector, FAISS, Pinecone, agents, tool calling, structured outputs, and LLM evaluation.
- Created explainable AI analytics with SHAP, Power BI, Seaborn, Plotly, and Matplotlib, translating model behavior into insights for business, compliance, and clinical stakeholders.
- Deployed secure cloud-native platforms on EC2, RDS, S3, Azure Functions, Key Vault, Synapse, Glue, API Gateway, Fargate, Kinesis, DynamoDB, CloudWatch, and Cosmos DB.
- Integrated RESTful model inference with PostgreSQL, MySQL, and SQL Server alongside tracing and observability, delivering 93% fraud precision while strengthening auditability and rollback.

TECHNICAL SKILLS

Languages	Python, SQL, Java, PySpark, Bash
ML & Deep Learning	Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, spaCy, SHAP, Autoencoders, Prophet, ARIMA, Classification, Forecasting, Anomaly Detection, Predictive Modeling
Generative AI & LLMs	Azure OpenAI, OpenAI API, LangChain, LlamaIndex, Hugging Face Transformers, Prompt Engineering, GPT, LoRA/QLoRA Fine-Tuning, vLLM, Model Quantization, LLM Evaluation
RAG & Vector Search	RAG, Azure AI Search, Vector Embeddings, Semantic Search, pgvector, FAISS, Pinecone, Agents, Tool Calling, Structured Outputs
Cloud Platforms	Microsoft Azure, AWS, Azure ML, Azure Data Factory, Databricks, Synapse, Azure Functions, Key Vault, S3, EC2, Lambda, Glue, RDS, API Gateway
Data & Streaming	Apache Spark, PySpark, Kafka, Delta Lake, Snowflake, Airflow, Azure Data Factory, Databricks, Event Hubs, Amazon Kinesis, ETL, Batch, Streaming
MLOps & DevOps	MLflow, Docker, Terraform, Azure DevOps, GitHub Actions, Jenkins, CI/CD, Model Monitoring, Experiment Tracking, Model Versioning, LLM Tracing, Observability
APIs & Microservices	REST APIs, FastAPI, Flask, Spring Boot, AWS Lambda, ECS, Fargate, API Gateway, SNS, SQS, Step Functions
Databases	PostgreSQL, MySQL, SQL Server, Amazon RDS, DynamoDB, Cosmos DB, ElastiCache Redis
Visualization & XAI	Power BI, SHAP, Seaborn, Plotly, Matplotlib
Security & Governance	Azure Key Vault, AWS IAM, VPC, KMS, PCI-DSS, HIPAA-Aligned Controls, Audit Logging, Model Governance
Testing & Quality	JUnit, PyTest, SonarQube, Git, CI/CD

PROFESSIONAL EXPERIENCE

Humana

AI/ML Engineer

Jan 2024 - Present

Healthcare payer | HIPAA

- Engineered high-risk patient classification with Azure Machine Learning using Python, SQL, and PySpark, improving detection accuracy by 34% while strengthening intervention workflow prioritization.
- Built secure healthcare pipelines using Azure Data Factory, Event Hubs, and Databricks, enabling scalable ingestion,

transformation, real-time analytics, and downstream processing.

- Delivered payer fraud models using XGBoost and LightGBM, integrating claim risk identification into review workflows and preventing inappropriate payment decisions.
- Developed Generative AI assistants using Azure OpenAI, LangChain, GPT summarization, and vector-based retrieval, accelerating longitudinal patient-history review by 50%.
- Automated patient intake using Azure Cognitive Services, OCR, and NLP, improving document processing, data consistency, compliance, and structured extraction.
- Created readmission forecasting with ARIMA and Prophet, providing capacity-planning analytics for staffing and resource allocation across regional facilities.
- Deployed real-time machine-learning APIs through FastAPI and Azure Functions, integrating patient health monitoring with clinical applications.
- Implemented RAG using Azure Cognitive Search and OpenAI over enterprise policies and clinical SOPs, enabling semantic retrieval and regulated knowledge discovery for physician queries.
- Established governed MLOps with MLflow, Azure DevOps, and Terraform, covering experiment tracking, model versioning, artifact management, rollback, HIPAA-aligned controls, and audit readiness.
- Enabled explainable AI using SHAP and Power BI so technical and business stakeholders could confidently evaluate model decisions.

Mastercard

Aug 2021 - Jul 2023

Software Engineer

Payments | PCI-DSS

- Engineered scalable transaction microservices using Java, Python, AWS ECS, Lambda, Spring Boot, API Gateway, and ElastiCache Redis, cutting system downtime 40%.
- Developed real-time fraud detection using Amazon Kinesis, Lambda, DynamoDB, NLP sentiment tagging, keyword extraction, and anomaly detection.
- Automated ETL reconciliation for CSV, JSON, and Parquet datasets using AWS Glue, Step Functions, and S3.
- Modernized legacy batch processing through SNS, SQS, Lambda, Kinesis, and event-driven serverless architecture, reducing execution time to 45 minutes.
- Established CI/CD automation using AWS CodePipeline, CodeBuild, Terraform, CloudFormation, and Fargate for repeatable, governed deployments.
- Secured PCI-DSS compliant cloud environments through IAM, VPC controls, KMS encryption, and audit logging, and tuned Amazon RDS SQL for performance.
- Implemented predictive CloudWatch alerting with dynamic thresholds, applying JUnit, PyTest, and SonarQube practices to improve incident management and code quality.
- Collaborated with AI/ML teams to expose fraud prediction models through REST APIs, enabling real-time scoring inside transactional flows.

EDUCATION

Master of Science, Computer Science

University of Texas at Arlington